

**GOVINDRAM SEKSARIA INSTITUTE OF
TECHNOLOGY & SCIENCE, INDORE
END SEMESTER EXAMINATION**

B.E. — First Year (Mechanical Engineering)

Maximum Marks: 56 Time: 3 Hours

SET B

Instructions:

- Attempt **ALL** questions.
- Diagrams must be neat and properly labelled.

UNIT I — Mechanical Properties of Materials (14 Marks)

- Q1.** (a) Define hardness. (2 Marks) (b) Define Poisson's ratio. (2 Marks)
Q2. Explain stress and strain with neat sketches. (3 Marks)
Q3. Explain Young's modulus, shear modulus and bulk modulus with formulas. (7 Marks)

UNIT II — Theory of Machines / Kinematics (14 Marks)

- Q4.** (a) Define mechanism. (2 Marks) (b) Define degree of freedom. (2 Marks)
Q5. Distinguish between slider-crank mechanism and four-bar mechanism. (3 Marks)
Q6. Explain the working principle of a four-bar chain with neat diagrams. (7 Marks)

UNIT III — Fluid Kinematics (14 Marks)

- Q7.** (a) Define streamlines. (2 Marks) (b) Define velocity potential. (2 Marks)
Q8. Explain rotational and irrotational flow. (3 Marks)
Q9. Derive Euler's equation of motion for an ideal fluid. (7 Marks)

UNIT IV — IC Engines & Automobiles (14 Marks)

- Q10.** (a) Define indicated power. (2 Marks) (b) Define brake thermal efficiency. (2 Marks)
Q11. Explain the working of a two-stroke engine with neat sketches. (3 Marks)
Q12. Draw Diesel cycle on P-V diagram and explain all processes. (7 Marks)